

# Day 1 - Facilitation Guide

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(2 hrs) ILT

### **I. Trainer & Students' Introductory Session - (10 Mins)**

- Trainer should introduce to students
- Students also will introduce themselves to the trainer
- **The trainer must identify the technical background of the students.**  
The trainer will try to elicit the background of the students (IT / Non-IT) to understand their current capabilities and decide on the pace of the session accordingly
- The trainer also declares all class rules and regulations to all students  
**Trainer will communicate the following rules:**
  - Students must attend the class regularly for 2 hours
  - Only informed leaves are allowed. Uninformed leaves will not be permitted and student will have to do self-study of missed topics
  - Laptop is mandatory for attending sessions. Mobile phones are strictly not allowed for attending sessions.
  - Exam leaves are permitted only once in the course duration and only on the day of the exam. Remaining days students need to attend classes
  - Student must complete all course work including tests, assignments, projects in order to be eligible for final assessment
  - Students must clear mock interview to be eligible for placements
  - In case any issue with internet, laptop, or any other issue leading to absence, student must keep the trainer informed

- Students must attend the class from a proper quiet location. Attending classes from road, car, train, open public places, etc is strictly prohibited
- **The sessions should happen in a camera-on mode.**  
Trainer and students both are expected to keep cameras on during the session. In case there are network issues, the camera can be turned off by the student. But when a question is asked, students must turn on the camera to answer.
- Trainer should explain the Module Objectives/WIIFM( What Is In It For Me)  
(Explain that they will learn Python programming language which is extremely popular till today and it will increase their employability.)

## **II. LMS & Diya App Orientation - (30 Mins)**

- LMS objective must be cleared
- LMS ID for students must be created
- Students must join by same device
- Diya App should be introduced by CM
- The session should be monitored by SME and DEAN

## **III. Introduction to Web Development**

Web development is the process of building and maintaining websites and web applications. It involves a combination of programming, design, and infrastructure management to create interactive and visually appealing online experiences.

## **IV. The Client-Server Model**

The client-server model is a foundational concept in web development. In this model, computers (or devices) are divided into two categories: clients and servers.

- **Clients:** These are typically web browsers, such as Google Chrome, Mozilla Firefox, or Safari, that users use to access websites and web applications.
- **Servers:** These are specialized computers that store and serve web content. They receive requests from clients, process those requests, and send back the requested resources, such as HTML pages, images, or data.

## **V. What is HTML?**

HTML, or HyperText Markup Language, is the standard language used to create and design documents on the World Wide Web. It's the backbone of most Web content and provides the basic structure for Web pages. HTML uses a system of markup tags to define elements within a document. These elements can include headings, paragraphs, links, images, forms, and other types of media.

Here are some key points about HTML:

**Markup Language:** HTML is a markup language, which means it uses markup tags to structure content on the Web. These tags are enclosed in angle brackets (< >) and typically come in pairs, with an opening tag and a closing tag. The content that falls between the opening and closing tags is what's displayed on the Webpage. HTML documents can contain both markup tag and plain text. We will explain this with an example in this session.

**Structure and Semantics:** HTML provides a way to define the structure and layout of a Webpage. It includes elements for headings, paragraphs, lists, tables, forms, and more. Additionally, HTML tags often carry semantic meaning, indicating the purpose of the content they enclose. For example, <h1> tags are used for main headings, <p> tags for paragraphs, and <a> tags for links.

**Hyperlinks:** HTML allows the creation of hyperlinks, which enable users to navigate between different Web pages. Links are created using the <a> (anchor) tag and the href attribute, specifying the URL of the destination page.

**Multimedia Integration:** HTML supports the embedding of various multimedia elements, such as images, videos, and audio files, into Webpages. This is achieved using tags like <img>, <video>, and <audio>.

**Forms:** HTML provides form elements like <input>, <textarea>, and <button> that allow users to interact with Webpages. Forms are commonly used for user input, such as submitting data or performing searches.

**Cross-Browser Compatibility:** HTML is a standardized language, ensuring that Web pages created with HTML can be displayed consistently across different Web browsers.

Developers often use HTML in conjunction with CSS (Cascading Style Sheets) and JavaScript to create visually appealing and interactive Web pages. CSS is used for styling and layout, while JavaScript is employed for enhancing interactivity and adding dynamic behavior to Web applications. Together, these technologies form the foundation of modern Web development.

## VI. HTML versions

HTML (HyperText Markup Language) has gone through several versions since its inception in the early 1990s. Each version introduced new features, improved existing ones, and aimed to enhance the capabilities of Web documents. Here's an overview of the major HTML versions:

**HTML 1.0 (1991):** The first version of HTML was introduced by Tim Berners-Lee, the inventor of the World Wide Web. It was a relatively simple language with a limited set of tags for structuring documents. The focus was on basic text formatting and linking.

**HTML 2.0 (1995):** This version added support for features like tables, form elements, text alignment, superscripts, and subscripts. HTML 2.0 also introduced the concept of browser support, ensuring consistency across different Web browsers.

**HTML 3.2 (1997):** HTML 3.2 brought significant improvements, including support for scripting languages (like JavaScript) and applets, text and table alignment options, and more image formatting features. It was widely adopted and helped establish many conventions still in use today.

**HTML 4.01 (1999):** HTML 4.01 introduced CSS (Cascading Style Sheets) support, enabling developers to separate content from presentation. It also included new structural elements, internationalization features, scripting support, and improved forms. HTML 4.01 remained the standard for a long time.

**XHTML 1.0 (2000):** XHTML (Extensible Hypertext Markup Language) was introduced as an XML-based reformulation of HTML 4.01. It aimed to make HTML more compatible with XML, which is a stricter and more structured markup language. XHTML 1.0 required well-formed documents and followed XML rules.

**HTML5 (2014):** HTML5 is the latest major version of HTML and introduced a wide range of new features and APIs. It includes native support for audio and video embedding, canvas for drawing, offline Web apps, improved forms, new parsing rules for enhanced flexibility, and more. HTML5 also brought semantic elements like <header>, <nav>, <article>, <section>, <aside>, and <footer>, allowing developers to create more meaningful page structures. Unlike XHTML, HTML5 is more forgiving of syntax errors, making it easier to write and maintain code.

HTML5 is the current standard for Web development and is supported by all modern Web browsers. It continues to evolve, with ongoing efforts to expand its capabilities and improve Web development practices.

HTML5 represents a significant upgrade over its predecessor, HTML, and has become the preferred choice for Web development today. HTML5 brings many advantages for both Web users and developers, including improved functionality, multimedia support, accessibility, faster page load times, easier development, enhanced performance, and mobile compatibility.

HTML5 has also opened up new possibilities for Web development, allowing developers to create more immersive, interactive, and engaging Web experiences. With the introduction of features like the canvas element, WebSockets, and the Web Audio API, developers now have the tools to create complex games, real-time communication apps, and other interactive applications directly in the browser.

## VII. HTML document structure

An HTML (HyperText Markup Language) document has a specific structure that consists of elements, tags, and attributes. Understanding this structure is fundamental to creating Webpages.

**Note:** The World Wide Web Consortium(**W3C**) is the main international standards organization for the World Wide Web. It was founded in 1994 and it develops standards and guidelines to help everyone build a Web based on the principles of accessibility, internationalization, privacy and security. **W3Schools** is a popular online platform that provides Web development tutorials and references for various Web technologies. To know about HTML tags and their usage please visit W3C sites.

Here's a breakdown of the basic HTML document structure:

### Document Type Declaration (DOCTYPE):

The DOCTYPE declaration defines the document type and version of HTML being used. It is not an HTML tag but rather an instruction to the Web browser about the version of HTML the page is using. For HTML5, the DOCTYPE declaration looks like this:

```
<!DOCTYPE html>
```

### <html> Element:

The <html> element is the root element of an HTML page. All other elements are nested inside it. It also contains the lang attribute to specify the language of the document:

```
<html lang="en">
  <!-- Other HTML elements go here -->
</html>
```

### <head> Element:

The <head> element contains meta-information about the HTML document, such as its title, character set, links to external stylesheets, scripts, and other metadata. It does not contain content visible to the user:

```
<head>
  <title>Page Title</title>
  <!-- Other meta-information and links to stylesheets/scripts go here -->
</head>
```

## **<body> Element:**

The <body> element contains the content of the HTML document that is visible to the user. This includes text, images, links, forms, and other elements that make up the Webpage:

```
<body>
  <h1>This is a Heading</h1>
  <p>This is a paragraph of text.</p>
  <!-- Other content elements go here -->
</body>
```

## **HTML Elements, Tags, and Attributes:**

Inside the <body> element, you can include various HTML elements to structure the content. Elements are represented by HTML tags, which have opening and closing tags. Opening tags contain the name of the element, and closing tags have a forward slash before the element name (e.g., <tag></tag>). Some HTML elements don't have closing tags and are self-closing (e.g., <img>).

Attributes provide additional information about elements and are written within the opening tag. For example, the href attribute in an anchor tag specifies the hyperlink destination:

```
<a href="https://www.example.com">Click me!</a>
```

## **Example of a Complete HTML Document:**

Putting it all together, a simple HTML document might look like this:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>My Webpage</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <h1>Welcome to My Webpage</h1>
  <p>This is a paragraph of text.</p>
  <a href="https://www.example.com">Click me!</a>
</body>
</html>
```

This example demonstrates a basic HTML document structure with the essential elements and tags. As you learn more, you can incorporate additional HTML elements and attributes to create more complex and interactive Web pages.

HTML tags are not case sensitive: That means `<P>` means the same as `<p>` in case of HTML. The HTML standard does not require lowercase tags, but **W3C** recommends lowercase in HTML, and demands lowercase for stricter document types like XHTML ( EXtensible HyperText Markup Language ). At W3Schools they use lowercase tag names for all their examples.

## VIII. Anatomy of an HTML tag

In HTML, a tag is a specific set of characters used to define elements within a document. Tags are fundamental to HTML syntax and are used to structure content on a Webpage. It consists of several parts, each serving a specific purpose. They consist of opening and closing parts, sometimes with additional attributes. Here's the breakdown of the anatomy of an HTML tag:

**Opening Tag:** An opening tag marks the beginning of an HTML element. It consists of the following parts:

**Tag Name:** The name of the HTML element. For example, in `<p>`, the tag name is "p", representing a paragraph element.

**Attributes:** Optional attributes provide additional information about the element. Attributes are written within the opening tag and come in name/value pairs like `attribute_name="value"`. For instance, in `<a href="https://www.example.com">`, `href` is an attribute specifying the hyperlink destination.

**Closing Tag:** A closing tag marks the end of an HTML element. It has the same tag name as the opening tag but with a forward slash (/) before the tag name. For example, `</p>` is the closing tag for the paragraph element.

**Content:** The content of the HTML element is the information or elements nested between the opening and closing tags. For example, in `<p>Hello, World!</p>`, "Hello, World!" is the content of the paragraph element.

**Self-Closing Tags (Optional):** Some HTML elements don't have content and don't require a separate closing tag. They are self-closing and end with a forward slash before the closing angle bracket. For example, `<img>` is a self-closing tag used to embed images.

In HTML5, the forward slash is optional, so `<img>` is equivalent to `<img />`.

Example Tag: <a> (Anchor Element)

Let's break down the anatomy of an <a> tag used to create hyperlinks:

Opening Tag: <a  
Attribute: href="https://www.example.com"  
Content: Click me!  
Closing Tag: </a>

Putting it all together:

```
<a href="https://www.example.com">This is a link</a>
```

**Attribute Name**

```
<a href="https://www.example.com">This is a link</a>
```

**Opening Tag**

**Attribute Value**

**Closing Tag**

In this example, the <a> tag creates a hyperlink. The href attribute specifies the URL to link to, and "Click me!" is the visible text on the Webpage. Remember that different HTML elements have different attributes and purposes, so the structure of the tag can vary based on the element being used.

## IX. Basic HTML document with example

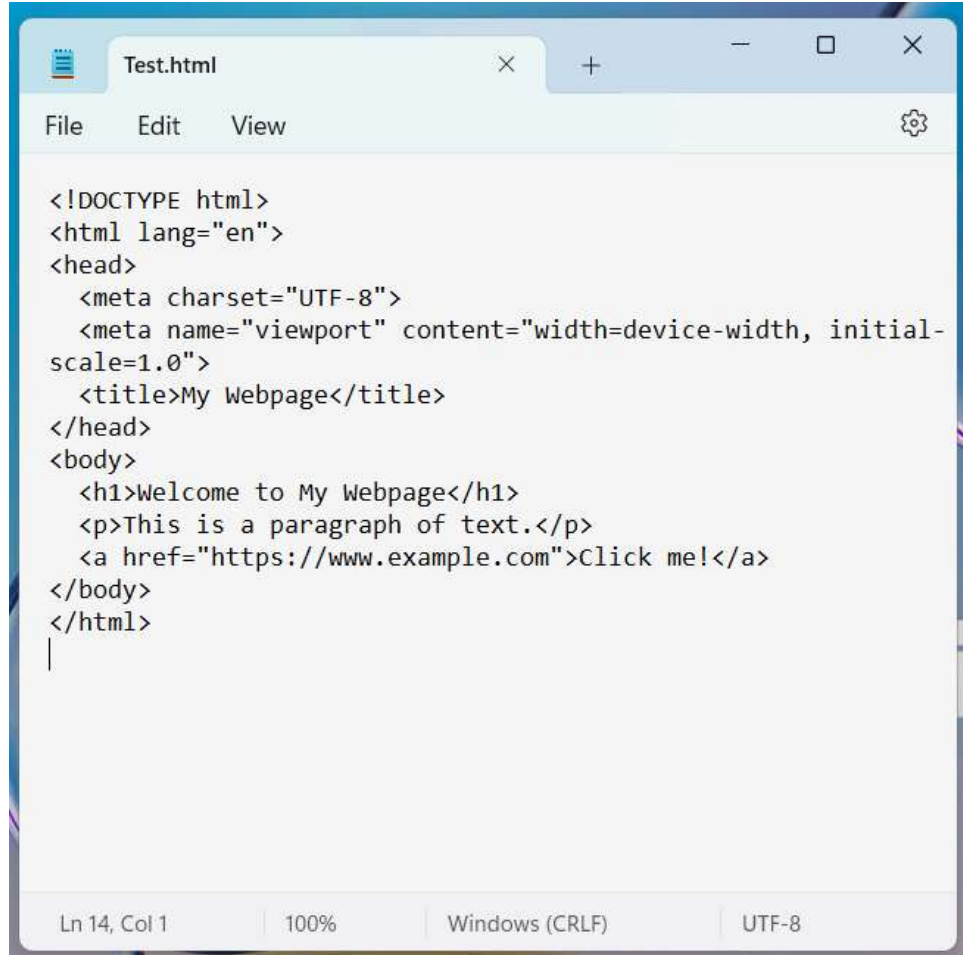
We have already discussed the structure and components of a basic HTML document. Now let us show you an example of a complete basic HTML document Putting it all together.

Create a file with name Test.html and add the following content:

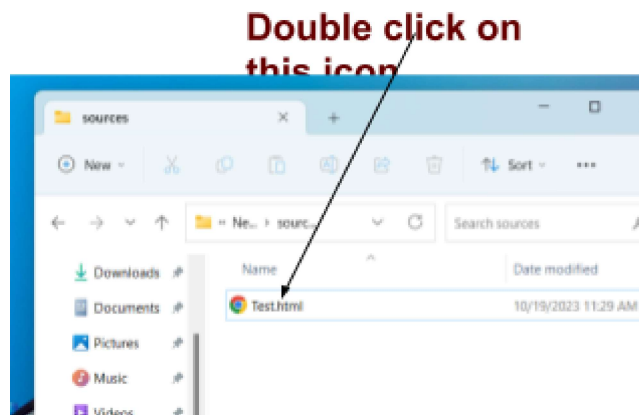
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>My Webpage</title>
</head>
<body>
  <h1>Welcome to My Webpage</h1>
```

```
<p>This is a paragraph of text.</p>
<a href="https://www.example.com">Click me!</a>
</body>
</html>
```

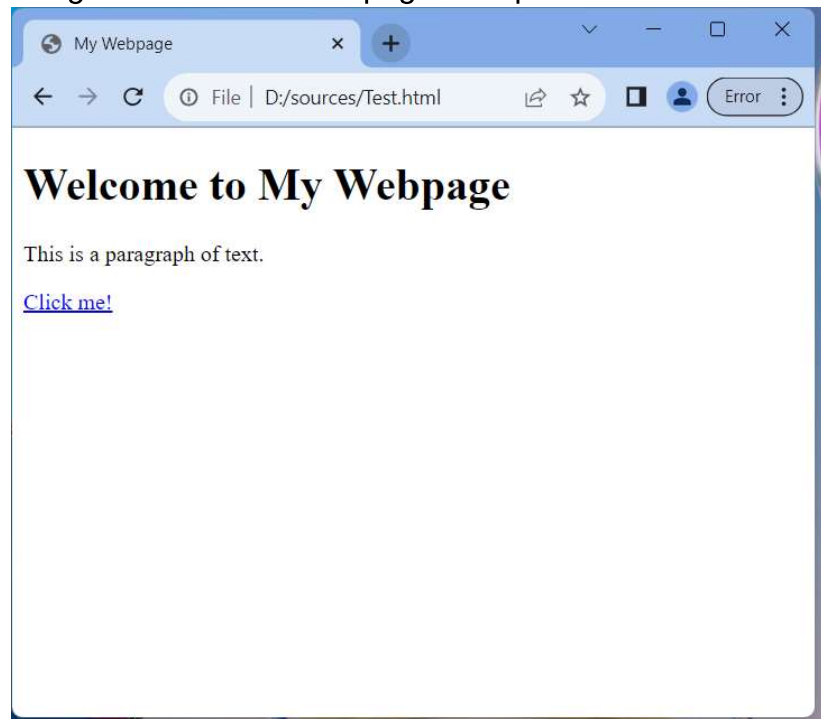
Save the file as Test.html in any directory.



Now open the windows file explorer and double click on the file icon.



After double clicking on the file icon the page will open in the Web browser.



In this example:

- `<!DOCTYPE html>` declares the document type and version (HTML5).
- `<html>` is the root element.
- `<head>` contains meta-information and the title of the document.
- `<body>` contains the visible content of the Webpage, including headings (`<h1>`), paragraphs (`<p>`), and a hyperlink (`<a>`).

When you save this code in an HTML file and open it in a Web browser, you'll see a Webpage with the title "My Webpage," a heading, a paragraph of text, and a clickable link. This is the basic structure of an HTML document.

## X. Examples of HTML usage in Real Life

HTML (HyperText Markup Language) is the backbone of the Internet and is used extensively to create Web pages and Web applications. Here are some real-life examples of how HTML is used:

**Websites:** Virtually all Websites on the Internet use HTML to structure their content. HTML is used to create headers, paragraphs, lists, forms, images, videos, links, and other elements that make up a Website.

**Blogs:** Blog platforms and content management systems rely heavily on HTML to present articles and multimedia content to readers.

**Online Stores:** E-commerce Websites use HTML to display product descriptions, prices, images, and shopping cart interfaces.

**News Websites:** News Websites use HTML to present news articles, images, videos, and interactive elements to readers.

**Emails:** HTML is used to create visually appealing and interactive email templates. HTML emails can include images, links, buttons, and styled text.

**Educational Platforms:** Online learning platforms use HTML to create course materials, quizzes, interactive lessons, and discussion forums.

**Web Applications:** HTML is used in Web applications to create user interfaces, input forms, dashboards, charts, and interactive elements.

**Government Websites:** Government Websites use HTML to provide information to citizens, including services, announcements, and official documents.

**Social Media Platforms:** HTML is used to create user profiles, posts, comments, images, videos, and other interactive elements on social media platforms.

**Documentation:** Technical documentation and user guides often use HTML to structure information, making it easy to navigate and read online.

**Online Forums and Communities:** HTML is used to structure discussion threads, user profiles, posts, and interactive features in online forums and community Websites.

**Personal Websites and Portfolios:** Many individuals use HTML to create their personal Websites, showcasing their work, skills, and portfolios.

**Web-Based Games:** HTML5, in combination with CSS and JavaScript, is used to create browser-based games and interactive experiences.

**Responsive Web Design:** HTML is used in responsive Web design techniques to create Websites that adapt and provide optimal viewing experiences across various devices and screen sizes.

**Web Forms:** HTML is used to create various types of forms, such as contact forms, registration forms, surveys, and feedback forms, enabling users to interact with Websites.

These examples illustrate the diverse range of applications for HTML in real-life scenarios, highlighting its central role in shaping the digital landscape.

## XI. Introduction to ChatGPT

### ChatGPT Exercise:

Please visit the site - <https://chat.openai.com/> and create your account if not yet created. Next, start with the first question on the prompt:

**I want to learn HTML. Could you suggest the best editors I can use to develop Webpages with HTML and CSS?**

I want to learn HTML. Could you suggest the best editors I can use to develop webpages with HTML and CSS?



Submit the query clicking the green arrow icon. Now go through the response received from the AI engine. Output will be something like the following screenshot.



Certainly! There are many great editors available for developing webpages with HTML and CSS. Here are some popular ones that you can consider:



1. **Visual Studio Code:** Visual Studio Code (VSCode) is a highly popular and feature-rich code editor developed by Microsoft. It has excellent support for HTML, CSS, and various other programming languages. VSCode offers a wide range of extensions that can enhance your web development experience.  
Website: [Visual Studio Code](#)
2. **Sublime Text:** Sublime Text is a lightweight and fast code editor with a sleek user interface. It's known for its speed and ease of use. Sublime Text supports HTML, CSS, and many other programming languages. It also has a vibrant plugin community.  
Website: [Sublime Text](#)
3. **Atom:** Atom is an open-source text editor developed by GitHub. It is highly customizable and has a rich ecosystem of packages and themes. Atom provides great support for HTML and CSS and allows you to collaborate with others using its Teletype feature.  
Website: [Atom](#)

Regenerate

***After the 2 hr ILT, the student has to do a 1 hour live lab. Please refer to the Day 1 Lab doc in the LMS.***